
The Floor Moves With the Probe: Trivial Baselines for Self-Supervised Learning on Satellite Imagery

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Abstract

1 Self-supervised objectives for satellite imagery are compared to one another and
2 almost never to a trivial baseline. We added two: patch-mean raw spectral bands,
3 which has zero parameters, and a random-initialised encoder. The conclusion
4 changed. On PASTIS crop segmentation, under a frozen probe with epochs, per-
5 step and effective batch, and probe budget matched across cells but compute
6 deliberately not matched, the floor is not a constant: the raw-band baseline moves
7 from 4.76 to 21.45 mIoU, a factor of 4.5, as the probe’s receptive field grows
8 from 1 to 9. The margin of the best objective over it moves with it. Temporal
9 latent prediction beats the raw-band floor by 10.56 ± 1.12 mIoU at receptive
10 field 1 and 4.78 ± 1.30 at 3, but by -0.15 ± 1.28 at 5, where the zero-parameter
11 baseline reaches 20.24 mIoU against the encoder’s 20.09, and -1.57 ± 1.28 at 9:
12 its entire advantage over a baseline with no encoder and no parameters is closed
13 by giving the probe a 5×5 convolution. Of the other four, three sit more than two
14 standard deviations below the binding floor and one is indistinguishable from it.
15 We therefore report rankings only at a stated probe capacity, with floors at that
16 same capacity. These are frozen-probe results at one training budget, and we report
17 the compute each cell used.

18 1 Introduction

19 Earth observation is natively temporal. The same location is imaged every few days, and crop type
20 is defined by a phenological trajectory rather than by the appearance of any single acquisition. The
21 objectives most often applied to satellite imagery, masked reconstruction and augmentation contrast,
22 operate on individual frames and discard the axis that carries the label. The natural alternative is a
23 temporal pretext task: predict the latent state of a location at a future acquisition from its past.

24 We set out to compare that objective against the standard ones on PASTIS [Sainte Fare Garnot and
25 Landrieu, 2021]. Our benchmark, like most in this literature, initially contained no trivial floor. There
26 was no random-initialised encoder and no raw-feature baseline. We added both, and the conclusion
27 changed.

28 Against patch-mean raw spectral bands, a baseline with no encoder and no parameters, four of the
29 five learned objectives we tested do not improve dense crop segmentation under a convolutional
30 probe. While checking that result we found something we had not anticipated: **the floor is not a**
31 **fixed quantity, and neither is any margin over it.** Sweeping the probe’s spatial receptive field from
32 1 to 9, the raw-band floor climbs from 4.76 to 21.45 mIoU while the best objective’s margin over it
33 falls from $+10.56$ to -1.57 . The advantage of the strongest self-supervised objective we tested over
34 a baseline with no parameters is closed entirely by enlarging the probe. A method evaluated at one
35 probe capacity with no floor reported at that capacity can appear to work while encoding little the
36 raw bands do not already carry.

Related work. Self-supervised pretraining for Earth observation is active, with models trained on large unlabelled satellite corpora [Cong et al., 2022, Wang et al., 2023]. Our concern is how they are evaluated. Corley et al. [2023] attribute apparent advantages of remote-sensing pretraining over ImageNet pretraining to preprocessing choices such as resizing and normalisation, and Tamazyan et al. [2025] report that the strongest remote-sensing foundation models do not outperform general-purpose models in-distribution. Both compare learned models against other learned models; we compare against no model at all, a weaker baseline and a more discriminating one. Predicting future latents from a causal context is likewise established: contrastive predictive coding [van den Oord et al., 2018] set the template and CF-JEPA [Lee and Sim, 2026] applies a mask-free forward-prediction JEPA to time series. We evaluate this family rather than propose it.

Contributions.

- We show that the value of a trivial floor, which floor binds, and the margin of any learned objective over it are all functions of probe capacity. Sweeping receptive field 1 to 9 moves the raw-band floor by a factor of 4.5 and takes the best objective’s margin from +9.4 to −1.2 across-seed standard deviations. Reporting a floor without its probe capacity is therefore not meaningful.
- At every capacity, we give each comparison as a margin in units of across-seed variance against a floor measured at that same capacity, so cells that merely tie with a zero-parameter baseline are not read as beating it.
- We evaluate a decision rule that was pre-registered before the floors were run, with a commitment to report the outcome either way.

2 Method

A series is $X \in \mathbb{R}^{T \times 10 \times 128 \times 128}$ with acquisition days $d \in [1, 366]^T$. A shared ViT tokenizes each frame into $N=256$ patches of width $D=512$ at patch size $P=8$, and a factorized temporal transformer aggregates frames 1: t causally. A narrower predictor ($D_p=384$) maps that context to the stop-graduated EMA-target embedding of frame $t+\Delta$. No frame after t enters the context, so the task carries no leakage; day-of-year enters as a periodic encoding. The objective is the established one [van den Oord et al., 2018, Bardes et al., 2024, Lee and Sim, 2026] and we restate it only to fix the configuration evaluated. Consecutive acquisitions are near-identical, so the task admits a degenerate constant solution; we apply a variance-covariance penalty [Bardes et al., 2022] to the context embedding ($\lambda_v=1.0$, $\lambda_c=0.04$). Zeroing both recovers pure I-JEPA [Assran et al., 2023] and collapses on this data (Section 4).

3 Experimental setup

PASTIS, official 5-fold split, 10 Sentinel-2 bands, 128×128 patches, validation fold. **We truncate each series to at most 32 acquisitions**, evenly spaced at evaluation and randomly subsampled during pretraining. PASTIS provides 38 to 61 per series; the truncation bounds encoder memory, which scales with $T \times N$ tokens, and applies identically to every cell including the floors.

Every trained cell shares the encoder architecture, the optimizer, 100 epochs, an effective batch of 192, and a probe budget of 15 epochs. Encoders are frozen and only heads train. We report three readouts: a 1×1 linear head, a light two-layer convolutional head, and parcel k -NN with $k=20$.

The two floors. `raw_features` has no encoder and no parameters: it averages the raw bands over each 8×8 patch, then takes the padding-masked mean over acquisitions, giving a **10-dimensional temporal-mean composite**. The probe reads 10 channels from it against 512 from every learned encoder. `random` is the temporal architecture with weights never updated, separating the objective from the inductive bias of temporal attention. Neither trains.

Batch configuration. Every cell in Table 1, SimCLR [Chen et al., 2020], BYOL [Grill et al., 2020] and MAE [He et al., 2022] included, was trained at per-step batch 12 with accumulation 16. Matching the per-step batch rather than only the effective batch matters here: SimCLR’s negative count and

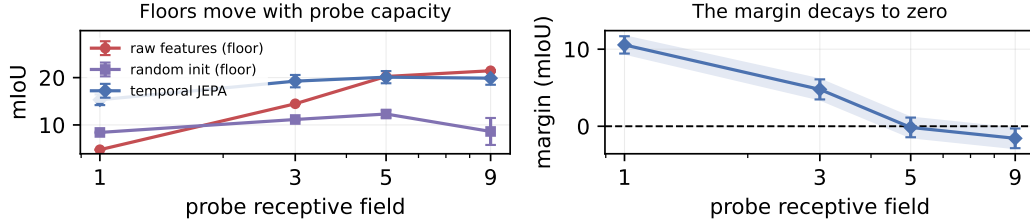


Figure 1: Neither the floor nor the margin over it is a property of the objective alone. **Left:** both floors and temporal JEPA as the probe’s spatial receptive field grows from 1 to 9. The raw-band floor climbs by a factor of 4.5 while the learned encoder flattens, and the two meet. **Right:** the same as a margin, the per-seed paired difference of temporal JEPA minus the raw-band floor, with a ± 1 sd band. It decays monotonically and reaches zero by receptive field 5. Receptive fields 1 and 3 are the published linear and conv columns, whose heads are structurally identical to the swept ones. Error bars and band are across-seed sd, $n=3$.

Table 1: PASTIS validation fold, frozen encoder, at probe receptive field 3 (conv) and 1 (linear). Mean $\pm$ sd, n per row; floors italic. Markers give the per-seed paired difference from that column’s binding floor in across-seed sd: $\downarrow$ below by ≥ 2 sd, $\approx$ within 2 sd. Last column is that ratio for conv. **All cells share per-step batch 12, effective batch 192, 100 epochs, 15-epoch probe;** compute is not matched (Section 3).

Objective	conv mIoU	linear mIoU	parcel k -NN	n	conv gap/sd
Temporal JEPA ($\Delta=1$)	19.26 $\pm$ 1.30	15.32 $\pm$ 1.13	68.67 $\pm$ 1.44	3	+3.7
<i>raw features (floor)</i>	<i>14.47 $\pm$ 0.01</i>	<i>4.76 $\pm$ 0.02</i>	<i>60.03 $\pm$ 0.52</i>	3	binds
<i>random init (floor)</i>	<i>11.15 $\pm$ 0.20</i>	<i>8.41 $\pm$ 0.15</i>	<i>60.58 $\pm$ 0.75</i>	3	−17.0
Spatial JEPA	13.90 $\pm$ 1.13 $\approx$	10.15 $\pm$ 1.65 $\approx$	58.92 $\pm$ 3.10 $\approx$	3	−0.5
SimCLR	7.96 $\pm$ 0.28 $\downarrow$	5.08 $\pm$ 0.02 $\downarrow$	56.84 $\pm$ 0.88 $\downarrow$	2	−24.2
BYOL	6.58 $\pm$ 0.89 $\downarrow$	4.96 $\pm$ 0.36 $\downarrow$	57.99 $\pm$ 0.44 $\downarrow$	2	−8.8
MAE	6.57 $\pm$ 2.46 $\downarrow$	4.79 $\pm$ 1.27 $\downarrow$	60.24 $\pm$ 4.52 $\approx$	3	−3.2

85 the variance-covariance term are both computed per micro-batch, so gradient accumulation does not
86 make the two equivalent.

87 **What is matched, and what is not.** Cells are matched on epochs, effective batch, per-step batch
88 and probe budget, **not on compute:** the temporal cells encode all 32 acquisitions while MAE encodes
89 one masked frame, so MAE uses 0.55 GPU-hours per seed against temporal JEPA’s 6.24. We answer
90 the equal-epoch question, not the equal-compute one; per-cell GPU-hours are in Appendix A.

91 **Pre-registration.** Before running the floors we committed, in a timestamped file, to the rule: the
92 claim is entitled iff temporal JEPA exceeds both floors and the margin over the larger floor exceeds
93 the across-seed standard deviation, with the outcome to be reported either way and the failure to lead
94 if it failed (Appendix D). The rule passed.

95 4 Results

96 **The pre-registered rule passes at the conv readout.** At the convolutional probe the rule was
97 written for, temporal JEPA exceeds the random-initialisation floor by 8.11 mIoU and the raw-band
98 floor by 4.78, the latter 3.7 times the across-seed sd. All three conditions hold, so the claim is entitled
99 under a rule fixed before the data was seen. Margins are means of per-seed paired differences taken
100 before rounding (Appendix D).

101 **The pre-registration inherited the choice this paper is about.** The rule fixed the convolutional
102 readout before we knew probe capacity was a free parameter. Evaluated across the sweep it passes at
103 receptive field 1 and 3 and fails at 5 and 9, where temporal JEPA does not exceed the raw-band floor
104 at all. We report it as passed at the capacity it was written for, and note that our own pre-registration

made the same unstated choice this paper is about: fixing a decision rule in advance does not help if the rule is stated at one capacity out of several.

The floor, and the margin over it, move with probe capacity. Our probe heads differ only in spatial receptive field: RF 1 is the strict linear probe and RF 3 the conv decoder used above, and we add RF 5 and 9 by stacking 3×3 convolutions. Over that range the raw-band floor climbs from 4.76 to 21.45 mIoU, a factor of 4.5, while temporal JEPA moves only from 15.32 to 19.88. The paired margin therefore falls monotonically: $+10.56 \pm 1.12$ at RF 1, $+4.78 \pm 1.30$ at 3, -0.15 ± 1.28 at 5 and -1.57 ± 1.28 at 9, or $+9.4$, $+3.7$, -0.1 and -1.2 across-seed sd (Figure 1). The margin is established at $\text{RF} \leq 3$ and absent at $\text{RF} \geq 5$. We do not claim a reversal at RF 9: at 1.2 sd the sign is not established. What is established is that the whole measured advantage of the strongest objective over a baseline with no parameters disappears once the probe can mix over 5×5 .

At the conventional capacity, one objective clears. Read at RF 3, temporal JEPA is the only objective whose margin over the binding floor exceeds two across-seed sd, on all three readouts. Spatial JEPA is within half an sd of the conv floor, one of the linear floor and seven tenths of the k -NN floor: it neither beats nor is beaten by a baseline with no encoder. MAE, BYOL and SimCLR are established below the binding floor on both dense readouts. The k -NN floors are 0.55 apart against a paired sd of 1.20, so we claim no reversal there.

Most of the margin is not from predicting the future. A shuffled-time control, identical in architecture and objective but with frames permuted and dates left chronological, keeps about 70% of the conv margin at seed 0 (18.38 against 20.03 and a 14.48 floor). These cells mostly buy use of the temporal axis, not prediction of the future (Appendix F).

The anti-collapse penalty is load-bearing. With $\lambda_v = \lambda_c = 0$ the representation collapses. Through the same pathway the probe uses, the effective rank¹ of the pooled embedding falls from 312.4 to 1.49 of 512 and off-diagonal covariance rises from 0.002 to 0.564 (Appendix H). Downstream, conv mIoU falls from 20.03 to 8.68 at seed 0, below both floors: without the penalty the objective is worse than not training. Both cells are single-seed, measured from the saved encoders.

Statistics. At $n=3$ no rank test can be informative, so we report margins in units of across-seed variance rather than p -values: the headline comparison is 19.26 ± 1.30 against 14.47 ± 0.01 . Appendix C gives the paired tests for readers who want them.

5 Discussion

The objective does shape the representation: decoding acquisition month from the frozen *spatial* features alone, temporal JEPA reaches 63.9% against spatial JEPA’s 47.0% at chance 8.3% (Appendix G). It is nonetheless not *worth* anything against raw bands once the probe can mix spatially, which is a claim about what the encoder adds over the input, not about whether it learned.

Reporting one probe without a floor at that same capacity therefore under-determines the conclusion, invisibly. We suggest reporting floors at every capacity used, and reading a margin as a property of the pair rather than of the objective. Our own headline is scoped accordingly: temporal latent prediction clears the binding floor on all three readouts *at receptive field 3*, not above.

6 Limitations

Seeds are $n=3$ except BYOL and SimCLR at $n=2$, whose sd is from two points and indicative only; the collapse ablation and shuffled-time control are single-seed. Our SimCLR uses the negative count its per-step batch affords, smaller than contrastive methods are usually tuned for, though MAE is not contrastive and fails comparably. Results are frozen-probe on one dataset, truncated to 32 acquisitions, without fine-tuning; U-TAE [Sainte Fare Garnot and Landrieu, 2021] is a ceiling, not a peer. The sweep varies receptive field at a fixed 15-epoch probe budget, so capacity and ease of optimisation are not separated.

¹erank = $\exp(-\sum_i p_i \log p_i)$, $p_i = \lambda_i / \sum_j \lambda_j$ over eigenvalues of the centred embedding covariance; 24,576 pooled tokens, validation fold.

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A Per-run results

Every run behind Table 1, ungrouped and unaveraged. All cells share per-step batch 12, gradient accumulation 16, effective batch 192, 100 epochs and a 15-epoch probe. Floors never train. An earlier pass at per-step batch 32 is superseded by these runs and is not pooled with them; the one result it changes is discussed in Section 4.

Exceptions, and one that did not survive. Spatial JEPA’s linear-probe mean, 10.15, is above both linear floors, though at 1.0 across-seed sd we do not claim the margin. A second apparent exception disappeared when we matched the configuration: in an earlier single-seed pass at per-step batch 32, BYOL scored 62.66 parcel k -NN and cleared both floors, but re-run at the matched batch over two seeds it scores 57.99 ± 0.44 , below both.

Results for the MAE, BYOL and SimCLR cells were read from the training logs rather than the result CSVs. The per-cell line that `run_matrix` prints after each probe is the same value it writes to the CSV, so no number is reconstructed or estimated. We read from the logs because those three cells ran as concurrent jobs against one shared CSV per seed, truncating some rows.

Cell	seed	conv mIoU	linear mIoU	parcel k -NN	GPU-h	configuration
BYOL	0	5.95	4.71	57.68	3.89	matched
BYOL	1	7.21	5.22	58.30	3.89	matched
MAE	0	4.77	3.92	55.19	0.44	matched
MAE	1	5.57	4.20	61.62	0.65	matched
MAE	2	9.37	6.24	63.90	0.56	matched
SimCLR	0	8.16	5.10	57.47	3.20	matched
SimCLR	1	7.77	5.07	56.22	3.23	matched
Spatial JEPA	0	14.84	11.58	60.17	0.98	matched
Spatial JEPA	1	14.20	10.52	61.20	0.96	matched
Spatial JEPA	2	12.65	8.34	55.39	0.83	matched
Temporal JEPA	0	20.03	16.03	67.84	6.19	matched
Temporal JEPA	1	19.99	15.91	70.33	6.29	matched
Temporal JEPA	2	17.75	14.02	67.84	6.25	matched
<i>random init</i>	0	11.35	8.26	60.79	0.00	floor, untrained
<i>random init</i>	1	11.15	8.55	61.20	0.00	floor, untrained
<i>random init</i>	2	10.95	8.43	59.75	0.00	floor, untrained
<i>raw features</i>	0	14.48	4.78	59.54	0.00	floor, untrained
<i>raw features</i>	1	14.47	4.75	59.96	0.00	floor, untrained
<i>raw features</i>	2	14.47	4.76	60.58	0.00	floor, untrained
Temporal JEPA, $\lambda_v=\lambda_c=0$	0	8.68	6.11	–	2.43	matched

B Probe-capacity sweep

Every point in Figure 1. Probe heads are n stacked 3×3 convolutions with hidden width 256 then a 1×1 classifier, giving receptive field $2n+1$; RF 1 is the 1×1 linear probe. RF 1 and RF 3 are the published linear and conv columns, whose heads are structurally identical to the swept ones, so the curve extends Table 1 rather than replacing it. Encoders are the same frozen checkpoints, re-probed. Probe head-init seed is 0 for every cell, matching the main runs. The last two columns are the mean per-seed paired difference of temporal JEPA from the raw-band floor and its ratio to the across-seed sd. $n=3$ throughout.

RF	raw features	random init	temporal JEPA	tjepa – raw	ratio
1	4.76 ± 0.02	8.41 ± 0.15	15.32 ± 1.13	$+10.56 \pm 1.12$	$+9.39$
3	14.47 ± 0.01	11.15 ± 0.20	19.26 ± 1.30	$+4.78 \pm 1.30$	$+3.67$
5	20.24 ± 0.02	12.31 ± 0.78	20.09 ± 1.29	-0.15 ± 1.28	-0.12
9	21.45 ± 0.14	8.62 ± 2.85	19.88 ± 1.39	-1.57 ± 1.28	-1.23

Which floor binds also moves with capacity: random initialisation binds at RF 1, raw bands at RF ≥ 3 . The random-initialisation floor is not monotone, peaking at RF 5 and falling at RF 9 with a large across-seed spread (± 2.85); we do not read anything into its shape. The raw-band floor is monotone and nearly seed-free, and it is the floor that binds at every RF above 1.

C Floor separation by readout

Whether the two floors are actually distinguishable, per readout. Paired across the three seeds, random minus raw.

Readout	raw features	random init	paired diff	t	separation
conv mIoU	14.47 ± 0.01	11.15 ± 0.20	-3.32 ± 0.20	-29.5	established, $17.0 \times$ sd
linear mIoU	4.76 ± 0.02	8.41 ± 0.15	$+3.65 \pm 0.16$	+39.3	established, $22.7 \times$ sd
parcel k -NN	60.03 ± 0.52	60.58 ± 0.75	$+0.55 \pm 1.20$	+0.80	not established

The reversal we claim is between the convolutional and linear readouts, where the floors swap by more than seventeen across-seed standard deviations in both directions. Under k -NN the floors are within half a standard deviation of one another, so we do not treat the nominal ordering there as a third reversal. It does not affect the conclusion: temporal JEPA and BYOL clear both k -NN floors, and spatial JEPA, SimCLR and MAE clear neither, so the set of objectives that clears is identical whichever floor is taken to bind.

At $n=3$ the Wilcoxon signed-rank test cannot fall below 0.125 one-sided regardless of effect size, so it is reported only to show it was run. The paired t -test is one-sided, justified by a hypothesis fixed before the floors were run, and summarises the margins rather than testing them independently. We regard the margin-to-seed-variance ratios as the primary evidence.

D Pre-registration

Committed to a timestamped file before the floor cells were run, reproduced verbatim.

P1, PASTIS: “temporal JEPA’s win reflects learning, not architecture.” *Entitled iff* `tjepa_h1` conv mIoU exceeds **both** random and `raw_features`, **and** the margin over the larger floor exceeds the across-seed standard deviation.

- If `tjepa_h1` $\leq$ random: the win is attributable to the temporal-attention *architecture*, not the objective. We will say so, and the headline claim is retracted to “the architecture helps.”
- If `tjepa_h1` $\leq$ `raw_features`: no representation learning is demonstrated on PASTIS at all.

The same file commits in advance to reporting each outcome, “PASS, FAIL, or INCONCLUSIVE . . . including any that contradict our prior claims,” and states that if P1 fails, the paper’s empirical section leads with that. The rule passed: $19.26 > 14.47 > 11.15$, with a margin of 4.78 against an across-seed standard deviation of 1.30.

E Hyperparameters

Identical for every trained cell. Encoder: patch size 8, embedding width 512, spatial depth 6, 8 heads, temporal depth 4, gradient checkpointing on. Predictor: width 384, depth 6, 12 heads. EMA momentum $0.996 \rightarrow 1.0$, linear schedule. Loss: L2 on LayerNormed targets, stop-gradient on the target branch, $\lambda_v=1.0$, $\lambda_c=0.04$. Optimiser: AdamW, lr 10^{-3} , 15 warmup epochs, cosine decay to 10^{-6} , weight decay cosine-ramped $0.04 \rightarrow 0.40$, 100 epochs, per-step batch 12 with accumulation 16, AMP on, horizontal and vertical flips. Data: 10 Sentinel-2 bands, 128×128 patches, at most 32 acquisitions per series, day-of-year positional encoding. Probe: 15 epochs, batch 8, both heads trained identically for every cell.

F Shuffled-time control

Separates “predicting the actual future helps” from “using the temporal axis helps”. The control is the `tjepa_h1` cell with one flag changed: each sample’s real frames are permuted while dates and `pad_mask` are left untouched, so the day-of-year encoding stays chronological and cannot reveal the permutation, and padding is never moved across the mask boundary. Architecture, objective, optimiser, epochs, batch and probe budget are otherwise identical, so a drop cannot be attributed to a different readout path.

	seed 0	conv mIoU	linear mIoU	parcel k -NN
263	temporal JEPA (real future)	20.03	16.03	67.84
	shuffled-time control	18.38	14.48	65.56
	<i>raw features</i> (floor)	14.48	4.78	59.54

264 Permuting the frames costs 1.65 conv mIoU, and the control still clears the raw-band floor by 3.90 of
 265 temporal JEPA's 5.55. The cost of shuffling is 1.3 across-seed sd of the temporal cell, so at $n=1$ we
 266 cannot establish it as non-zero. The control did not collapse: its effective rank rose through training
 267 to 411, comparable to the trained cell, so the small gap is not an artifact of a degenerate run.

268 G Mechanism probe

269 Acquisition time decoded from the frozen *spatial* features. `encode_full`, the per-frame spatial
 270 ViT, is probed; the temporal pathway is not, because it carries an explicit day-of-year encoding and
 271 probing it would be circular. Validation fold, seed-0 checkpoints, 4800 training and 4800 evaluation
 272 frames, ridge probes for both tasks.

	encoder	month accuracy (chance 8.3%)	DOY circular MAE
273	Temporal JEPA	63.9%	29.9 days
	Spatial JEPA	47.0%	40.8 days

274 Single seed. This measures how much seasonal information the spatial representation carries, not
 275 downstream accuracy, and is reported as supporting evidence rather than as a headline result.

276 H Collapse diagnostics

277 The per-step diagnostics printed during training were not retained, so the values in Section 4 were
 278 measured after the fact from the saved encoders. Both cells were processed identically: load the
 279 encoder checkpoint, run the validation fold through `encode_temporal`, apply the padding-masked
 280 temporal mean, and pool over patches, which is the same pathway the frozen probe uses for JEPA
 281 cells. This measures the representation that is actually evaluated rather than the training-time context
 282 embedding. Statistics are over 24,576 token embeddings.

		VICReg on	VICReg off
283	per-dimension std	0.948	0.172
	effective rank (of 512)	312.4	1.49
	off-diagonal covariance	0.0025	0.564
	conv mIoU (seed 0)	20.03	8.68

284 Both are single-seed, at seed 0, under the matched configuration.